Due Date

You must *submit* the source code for the solution to this lab exercise to *Moodle* by

Monday, October27, 2025

in order to receive full credit for this work. You must also *demonstrate* the solution to the instructor during class, at the earliest opportunity.

Lab Exercise

Write a java program containing three classes: **MyPoint**, **NewPoint,** and **Lab11a**. The **NewPoint** class must be a subclass of **MyPoint**. The **Lab11a** class must create objects of the **NewPoint** class.

The MyPoint class:

An object of the **`MyPoint`** class represents a point with **`x`**- and **`y`**-coordinates. The **`MyPoint`** class must be in a file named **`MyPoint.java`**. The **`MyPoint`** class must contain the features listed below:

`double x, y;`	Data fields (variables) that represent the coordinates of the point.
`double getX()`	An accessor method ("getter method") that returns the **`x`** coordinate of the point.
`double getY()`	An accessor method ("getter method") that returns the **`y`** coordinate of the point.
`MyPoint()`	A no-arg constructor that creates a point **`(0,0)`**, with **`x`** and **`y`** both set to zero.
`MyPoint(double xValue, double yValue)`	A constructor that constructs a point with the specified coordinates.
`double distance(double xValue, double yValue)`	A method named **`distance`** that returns the distance from **`this`** point to another point with the specified **`x`**- and **`y`**-coordinates.
`double distance(MyPoint otherPoint)`	A method named **`distance`** that returns the distance from **`this`** point to another object of the **`MyPoint`** type.

The NewPoint class:

Design and implement a class named **`NewPoint`** that extends the **`MyPoint`** class.

The **`NewPoint`** class must contain:

- A private, static variable called **`numberOfObjects`** that contains a count of how many **`NewPoint`** objects have been created, since the program started.
- A no-arg constructor that creates a point **`(0,0)`**, with **`x`** and **`y`** both set to zero.

- A constructor that constructs a point with specified coordinates:

  `NewPoint(double xValue, double yValue)`

- A static method called **`getNumberOfObjects`** that returns the current value of the static **`numberOfObjects`** variable.

The Lab11a class:

Write a class named **`Lab11a`**, that contains a **`main()`** method. The **`Lab11a`** class must be in a file named **`Lab11a.java`**. The **`main()`** method must contain a loop that repeats the following steps until the user enters an input value of "**`q`**":

- Prompt the user to input the coordinates of two points. The **x** and **y** values can be input on the same line, separated by a space character. (HINT: use the **`split()`** method of the **`String`** class.)
- Create a **`NewPoint`** object for each pair of **x** and **y** values.
- Display the distance between them, using the **`System.out.printf()`** method. (This will enable precise control of the formatting of the output text.)
- Display the current value of the **`numberOfObjects`** variable.

Sample Output

The output for the program should look like the following:

```
Enter x and y coordinates for p1 (or q to exit): 0 0
Enter x and y coordinates for p2 (or q to exit): 3 4
p1=( 0.00000, 0.00000), p2=( 3.00000, 4.00000), numberOfObjects=2
Distance (object to object)          =  5.00000
Distance (object to X,Y coordinates) =  5.00000
Enter x and y coordinates for p1 (or q to exit): -1 -1
Enter x and y coordinates for p2 (or q to exit): 2 3
p1=(-1.00000,-1.00000), p2=( 2.00000, 3.00000), numberOfObjects=4
Distance (object to object)          =  5.00000
Distance (object to X,Y coordinates) =  5.00000
Enter x and y coordinates for p1 (or q to exit): 10 20
Enter x and y coordinates for p2 (or q to exit): 50 50
p1=(10.00000,20.00000), p2=(50.00000,50.00000), numberOfObjects=6
Distance (object to object)          = 50.00000
Distance (object to X,Y coordinates) = 50.00000
Enter x and y coordinates for p1 (or q to exit): q
```